

 YOUNG INNOVATOR PROGRAMME (YIP)

SPANDANA

A System to Detect How Plants
Communicate Stress in Real Time

PRESENTED BY

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SYSTEM STATUS: ACTIVE

CURRENT LIMITATIONS

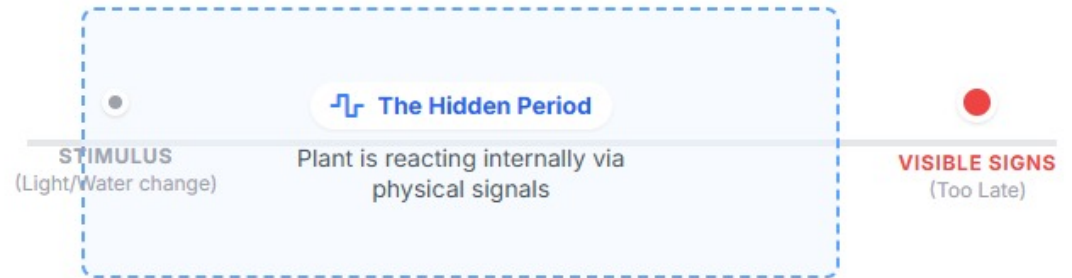
The Missing Window

Why traditional observation misses the critical moment of response.

👁 Visible Late Signs

- Leaves drooping or wilting
- Visible colour changes
- Growth rate slowing down

"By the time we see these, the plant has been stressed for a long time."



Our Project Goal

To access this hidden period and observe the reaction **as it happens**.

Shift in Thinking



MOST STUDIES FOCUS ON

"The Internal Chemistry"



Chemical Markers

Hormones, enzymes, molecular changes



Gene Expression

Transcriptomics and lab analysis



Visible Traits

Late-stage physical symptoms

Result: Accurate but slow & often destructive



WE ASKED A NEW QUESTION

physical signal?



Real-Time Physics

Not biology textbooks, but immediate physical behavior



Non-Invasive Signal

Observing without destroying the sample



Continuous Monitoring

Like a heart monitor for plants

What does the plant do immediately?

When something affects a plant—like a change in environment—reaction is inevitable.



LIGHT



WATER



DARKNESS



TOUCH

- ⌚ Before it wilts.
- 🌿 Before it changes colour.
- 👁️ Before anything becomes visible.



DEFINITION

What We Mean by 'Communication'

It is not about language; it is about internal physiological signaling.

✗ WE DO NOT MEAN:

Audible Sound



Language

✓ WE DO MEAN:**Internal Signals**

Information traveling through the stem.

**Physical Responses**

Micro-movements and pressure shifts.

**Changing Patterns**

Behavior that shifts under stress.

ANALOGY In Animals

A heartbeat changes rhythm under stress or fear.

**🌿 In Plants**

Stem vibrations change pattern under environmental stress.



Core Idea: Stress Becomes Physical

"Any stress creates energy inside a living system.
That energy has to go somewhere."



1. The Stimulus

ENVIRONMENT CHANGE

Light change, water deficit, physical touch,
or temperature shift affects the plant.



2. Energy Buildup

THERMODYNAMIC RESPONSE

The plant's internal system generates
potential energy as it attempts to maintain
homeostasis.



3. Physical Release

OUR FOCUS AREA

Energy dissipates as mechanical work:

- ✓ Internal pressure shifts
- ✓ Tissue contraction
- ✓ Stem Micro-vibrations



Conclusion: Plants may be **"talking"** through very small physical signals.

METHODOLOGY

Our Approach: Observe Directly



Instead of observing the environment, we observe the **plant body** directly.



The Stem as Carrier

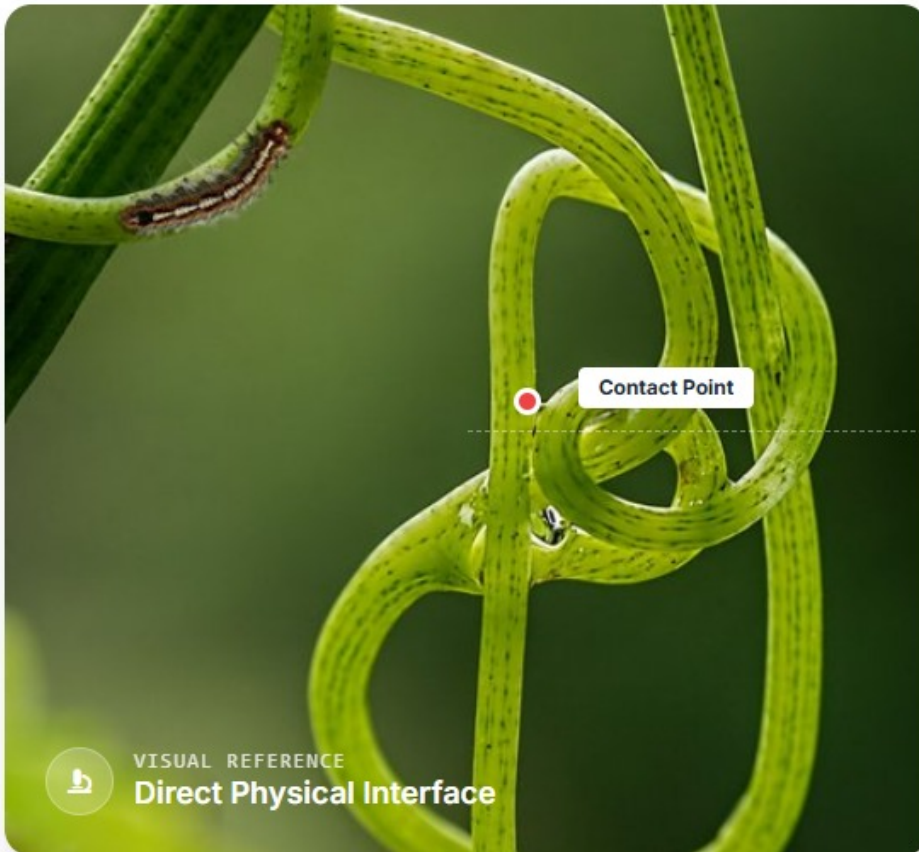
We treat the plant stem as a physical signal carrier, transmitting internal mechanical vibrations.



Reflecting Response

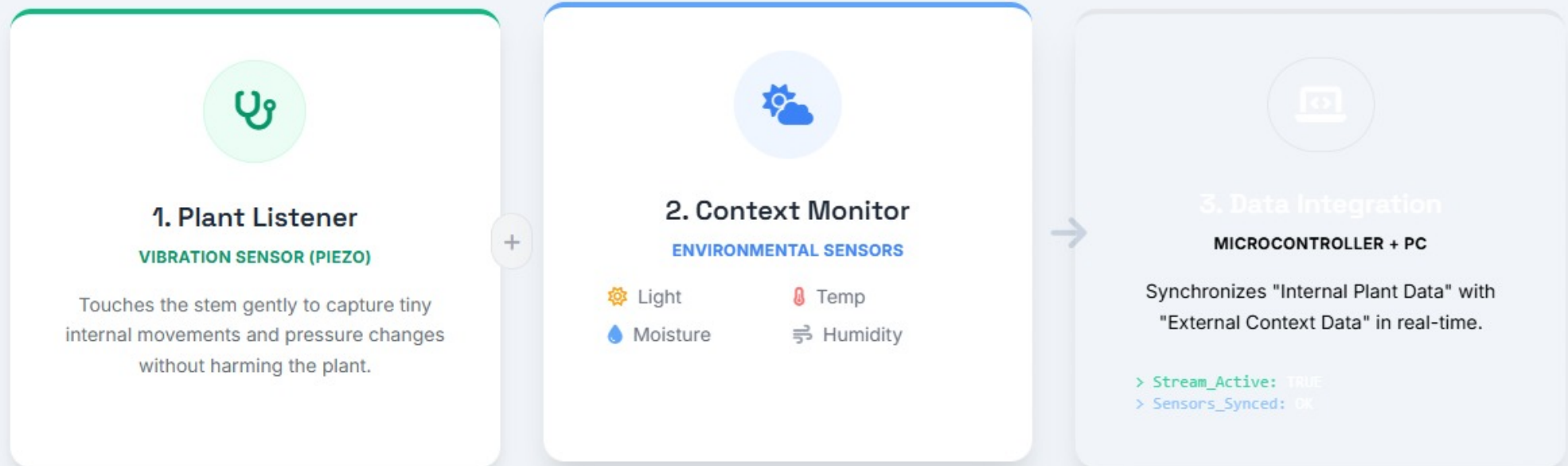
If the plant responds internally to stress, the stem structure physically reflects that change instantly.

"Listening to the source, not just the surroundings"



The System (Simple View)

We built a small, non-damaging system. The **vibration sensor listens to the plant**, while other sensors give context to what is happening around it.



The system is designed to be low-cost, portable, and non-destructive.

What Others Usually Do



MONITORING THE SURROUNDINGS

"External Parameters"



Light Levels

Intensity, spectrum, duration



Temperature & Humidity

Ambient air conditions



Soil Moisture

Water availability in the root zone

Result: We know the **Context**, but not the **Plant**.



THE MISSING INSIGHT

experiencing?



Hidden Response

Environment data doesn't reveal internal stress



Subjective Experience

Two plants in the same heat may react differently



Delayed Awareness

We only see trouble when it's physically visible

"Condition" ≠

Continuous Real-Time Listening

LIVE STREAMING DATA

● RECORDING ACTIVE

The system streams data continuously to a central computer, allowing us to observe the plant's temporal behavior—identifying **quiet periods**, **sudden bursts**, and **repeating patterns**.

24/7
MONITORING**50Hz**
SAMPLE RATE**ms**
LATENCY

Why Vibration?



THE PROBLEM WITH

"Airborne Audio"



Environmental Noise

Captures wind, traffic, and human sounds



Signal Dilution

Tiny internal sounds are lost in open air



False Positives

Hard to distinguish plant vs. external sources

Result: High interference & poor data fidelity



OUR SOLUTION

Sensing

Vibration



Precision Capture

Detects tiny internal movements directly at the source



Noise Immunity

Contact sensor ignores ambient air noise



Non-Invasive

Safe attachment without harming plant tissue

How We Detect a Response



1



Feature Extraction

The raw signal is analyzed for sudden shifts in energy and frequency content. We look for **deviations from silence**.

2



Anomaly Scoring

We calculate a deviation score (Z-Score). Is the change **statistically significant** compared to the plant's recent baseline?

3



Cluster Confirmation

True biological responses aren't single spikes. We confirm a "response" only if anomalies occur in **dense temporal clusters**.

VISUALIZATION OF LOGIC

● Noise — Adaptive Threshold ● Confirmed Response



No Fixed 'Normal': Adaptive Baselines

"Plants don't behave the same all the time.
The system learns what 'normal' looks like dynamically."



1. Variable Reality

THE CHALLENGE

Plant activity changes naturally with day/night cycles and humidity. A static number (e.g., "50") would cause false alarms.



2. Sliding Window

THE SOLUTION

The system continuously observes the last N minutes of data to calculate a "Moving Average" baseline.



3. Dynamic Detection

THE RESULT

Responses are detected relative to the *current* context. "High" at night is different from "High" at noon.



This approach avoids false positives and allows detecting **subtle anomalies** regardless of the time of day.

Key Signal Behaviors Observed

EXPERIMENTAL DATA
● PATTERN RECOGNIZED

Contrary to random noise, plant signals displayed distinct behavioral states. We observed long **quiet intervals** punctuated by **high-energy bursts** and structured **repeating motifs**—characteristics typical of biological communication, not static interference.

>90%
SIGNAL
CLARITY

3
DISTINCT
STATES

**Non-
Random**
DISTRIBUTION



VALIDATION EXPERIMENT

Darkness Test: Internal Origin

Even when **light was fully removed**, the plant's signal activity continued uninterrupted.



Signals Persist

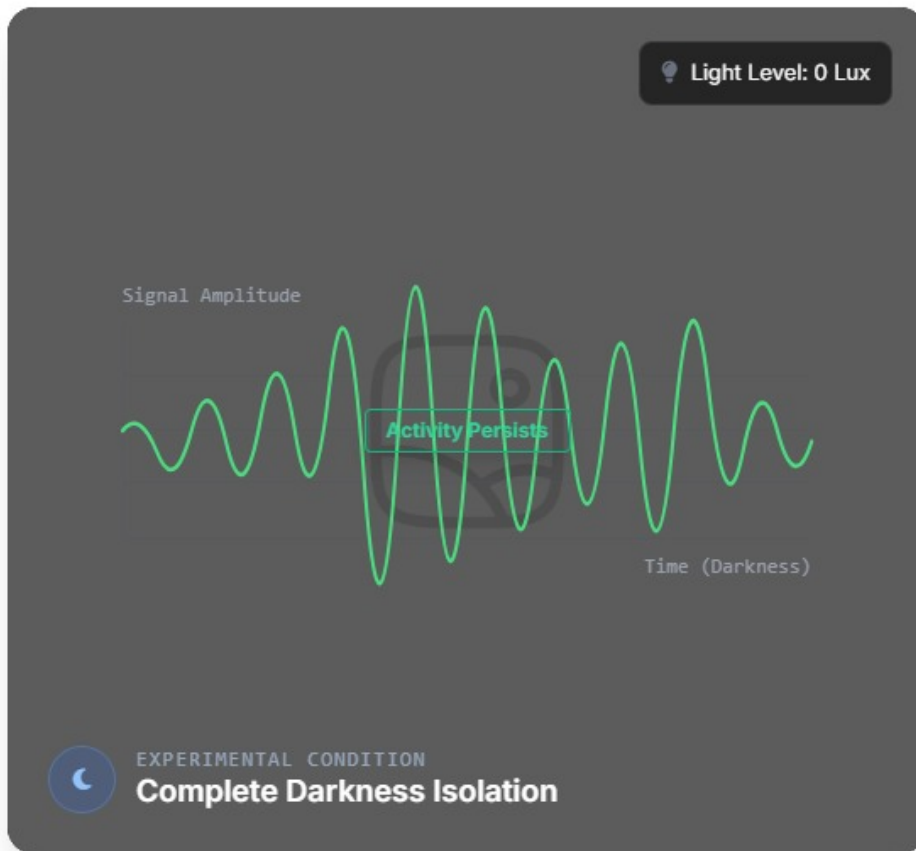
Internal responses were still present in the dark, proving these are not merely photovoltaic reactions to light.



Not Electronics Noise

This confirmed that signals originate **from inside the plant**, distinct from environmental light artifacts.

"The plant is active, even in the dark."



Experiments Conducted

 Non-Destructive Testing

Normal / Healthy

BASELINE CONTROL

Plants observed under optimal conditions to establish baseline vibration patterns and metabolic noise.

● 24h Duration



Water Deficit

DROUGHT STRESS

Gradual reduction of soil moisture to observe transition from healthy to stressed signaling states.

● 3-Day Cycle



Sudden Light

PHOTOPERIOD SHOCK

Abrupt changes in light intensity to trigger immediate photosynthetic and stomatal responses.

● Instant Stimulus



Complete Dark

SIGNAL VALIDATION

Observation in total darkness to rule out photo-electronic interference and confirm internal origin.

● 12h Isolation



Touch / Wind

MECHANICAL STIMULUS

Controlled gentle physical disturbance to calibrate the sensor against external mechanical noise.

● Event-Based



All experiments were conducted without harming the plant (Non-Invasive).

FINDINGS

CORE CONCLUSION

HYPOTHESIS
Confirmed

What This Shows



Measurable Signals

Plants produce distinct, measurable physical signals when experiencing environmental change, long before any chemical analysis would typically be done in the field.



Pre-Symptomatic

These signals appear **before** visible symptoms like wilting or yellowing. This creates a new window for early intervention.



Active Responders

Plants are not passive objects. They are actively responding systems, constantly adjusting their internal physical state to survive stress.

 SPANDANA Detects Here

THE EARLY DETECTION ADVANTAGE



T = 0

Stress Event



HOURS / DAYS

Internal Physical Signal



LATE STAGE

Visible Symptoms

(Wilting, Color Change)

**Hidden Phase:** Only detectable via direct monitoring.**Visible Phase:** Traditional observation methods start here (often too late).

Water Stress: Persistent Adjustment

LONG DURATION OBSERVATION

STRESS INDUCED

Under reduced water availability, signals did not decay quickly. Instead, we observed **persistent repeating patterns** over long durations, suggesting ongoing internal physiological adjustment rather than a single acute reaction.

48h

DURATION

High

PERSISTENCE

>150

EVENTS/HR



SIGNIFICANCE

PARADIGM SHIFT

Why This Matters



IMPACT

New Perspective



Reframing Perception

It changes how we think about plants. Instead of passive objects, we begin to see them as responsive, communicating living systems with their own internal experiences.



Proactive Care

We move from "autopsy" (diagnosing damage after it happens) to "empathy" (understanding needs in real-time), enabling better care and research.



New Science

It opens a new way to study plant physiology using physical signal patterns, complementing existing chemical and genetic approaches.

THE OLD QUESTION



"What happened to the plant?"

Reactive. Focusing on visible damage, symptoms, and past events.

Past Tense



THE SPANDANA QUESTION



*"What is the plant experiencing **right now**?"*

Proactive. Focusing on internal state, immediate signals, and real-time needs.

Present Tense

Study *Limitations*



STATUS

Early Exploration

🔗 Experimental Constraints

Physical and biological boundaries of our current testing phase.



Limited Species Scope

Tests focused on a small number of specific plants (e.g., Balsam). Results may not yet generalize to all plant types.



Environmental Controls

Conducted in school laboratory settings. Future work requires sterile climate chambers to eliminate all variables.



📊 Analytical Gaps

Areas where our data processing methodology needs expansion.



Basic Signal Classification

Current system uses threshold-based patterns. We have not yet implemented advanced AI/ML for signal sorting.



Replication Needs

Sample size was small. Large-scale controlled studies are needed to establish statistical significance.

🔍 Transparency: This is a proof-of-concept study. We acknowledge these limitations as the roadmap for future improvement.



What We Are *Not* Claiming

✓ What We ARE Doing

Our project focuses strictly on signal acquisition and pattern recognition.

✓ Listening System

Building a tool to detect physical micro-vibrations that were previously unnoticed.

✓ Pattern Observation

Identifying that patterns change under stress conditions.



✗ What We are NOT Claiming

We maintain strict scientific boundaries to avoid overstatement.

✗ Specific Diagnosis

We cannot say "This plant has fungal infection X". We only detect "Stress is present".

✗ Biological Mechanism

We do not explain the exact cellular biology causing the vibration. That is for biologists.

✗ Outcome Prediction

We cannot predict if the plant will die or recover. We only monitor current state.

— FINAL MESSAGE —

“ Plants are alive. Living systems respond.

We just started **listening.**”

This project shows that plants are constantly communicating internally — through signals too small for us to notice naturally, but vital to their survival.

—  —
Thank You

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•  Don Bosco School, Liluah

•  3/1/2026



Future Directions

Moving from a **proof-of-concept** to a comprehensive plant communication platform. Our goal is to decode the language of plant stress.



STEP 1

Species Comparison

Expanding beyond current test subjects. Do monocots and dicots signal differently?

Varieties

Stem Types



STEP 2

Systematic Stimuli

Controlled testing of specific stressors: Salinity, pathogen mimics, and temperature extremes.

Stressors

Controls



STEP 3

AI Pattern Recognition

Training Machine Learning models to classify signal types automatically without human review.

Classification

Neural Nets



ULTIMATE GOAL

Response Library

Building a "dictionary" of plant signatures. A universal reference for plant stress signals.

Database

Open Access



TIMELINE

Next 12-18 Months



OUTCOME

Standardized Monitoring Protocol

"From listening to understanding"[VIEW DETAILED PLAN](#)

A Multi-Sensor, Non-Invasive System for Real-Time Detection and Classification of Plant Stress Signals

Exploring Real-Time Physical Stress Responses in Plants



fig 1.0 – Signal Acquisition Concept

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